

ALFAAS22081

## Manganese(II) sulfate tetrahydrate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | Manganese(II) sulfate tetrahydrate<br>Manganese(II) sulfate tetrahydrate                                                                                                                                                                                           |
| Cat No. :                     | S22081                                                                                                                                                                                                                                                             |
| Synonyms                      | Sulfuric acid, manganese(2+) salt (1:1), tetrahydrate.                                                                                                                                                                                                             |
| CAS No                        | 10101-68-5                                                                                                                                                                                                                                                         |
| Molecular Formula             | MnSO4.4H2O                                                                                                                                                                                                                                                         |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                          |                         |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------|
| Physical State<br>Solid                                                                                                                                                                                  | Appearance<br>Light red | Odor<br>Odorless |
| <b>Emergency Overview</b><br>May be harmful if swallowed. Causes serious eye damage. May cause damage to organs through prolonged or repeated exposure. Toxic to aquatic life with long lasting effects. |                         |                  |

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Acute Oral Toxicity                                  | Category 5 |
| Serious Eye Damage/Eye Irritation                    | Category 1 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Chronic aquatic toxicity                             | Category 2 |

#### Label Elements



**Signal Word****Danger****Hazard Statements**

H303 - May be harmful if swallowed

H318 - Causes serious eye damage

H373 - May cause damage to organs through prolonged or repeated exposure

H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May be harmful if swallowed. May cause damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Toxic to aquatic life with long lasting effects. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                          | CAS No     | Weight % |
|------------------------------------|------------|----------|
| Manganese(II) sulfate tetrahydrate | 10101-68-5 | >95      |
| Manganese(II) sulfate              | 7785-87-7  | -        |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

Causes severe eye damage. Causes eye burns.

**Self-Protection of the First Aider**

Use personal protective equipment as required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Ensure adequate ventilation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component                          | ACGIH TLV                                                 | OSHA PEL                                                               | NIOSH                                                                                | The United Kingdom                                                                                                                               | European Union                   |
|------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Manganese(II) sulfate tetrahydrate | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | (Vacated) Ceiling: 5 mg/m <sup>3</sup><br>Ceiling: 5 mg/m <sup>3</sup> | IDLH: 500 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr | TWA: 0.05 mg/m <sup>3</sup> (8h) |

**SAFETY DATA SHEET****Manganese(II) sulfate tetrahydrate**

|                       |                                                           |                                                                        |                                                                                      |                                                                                                                                                  |                                  |
|-----------------------|-----------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Manganese(II) sulfate | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | (Vacated) Ceiling: 5 mg/m <sup>3</sup><br>Ceiling: 5 mg/m <sup>3</sup> | IDLH: 500 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr | TWA: 0.05 mg/m <sup>3</sup> (8h) |
|-----------------------|-----------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water

system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                      |                                          |
|------------------------------------------------|--------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Light red                            |                                          |
| <b>Physical State</b>                          | Solid                                |                                          |
| <b>Odor</b>                                    | Odorless                             |                                          |
| <b>Odor Threshold</b>                          | No data available                    |                                          |
| <b>pH</b>                                      | 4.5-6.0                              | 50g/L (20°C)                             |
| <b>Melting Point/Range</b>                     | 26 - 27 °C                           |                                          |
| <b>Softening Point</b>                         | No data available                    |                                          |
| <b>Boiling Point/Range</b>                     | No information available             |                                          |
| <b>Flash Point</b>                             | No information available             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                       | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available             |                                          |
| <b>Explosion Limits</b>                        | No data available                    |                                          |
| <b>Vapor Pressure</b>                          | No data available                    |                                          |
| <b>Vapor Density</b>                           | Not applicable                       | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available                    |                                          |
| <b>Bulk Density</b>                            | No data available                    |                                          |
| <b>Water Solubility</b>                        | Soluble in water                     |                                          |
| <b>Solubility in other solvents</b>            | No information available             |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                      |                                          |
| <b>Autoignition Temperature</b>                | No data available                    |                                          |
| <b>Decomposition Temperature</b>               | No data available                    |                                          |
| <b>Viscosity</b>                               | Not applicable                       | Solid                                    |
| <b>Explosive Properties</b>                    | No information available             |                                          |
| <b>Oxidizing Properties</b>                    | No information available             |                                          |
| <b>Molecular Formula</b>                       | MnSO <sub>4</sub> ·4H <sub>2</sub> O |                                          |
| <b>Molecular Weight</b>                        | 223.05                               |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                           |
|---------------------------------|-----------------------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions.                           |
| <b>Hazardous Reactions</b>      | None under normal processing.                             |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                  |
| <b>Conditions to Avoid</b>      | Incompatible products. Excess heat. Avoid dust formation. |
| <b>Materials to avoid</b>       | No information available.                                 |

Hazardous Decomposition Products Sulfur oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

Product Information See actual entry in RTECS for complete information.

## (a) acute toxicity;

| Component                          | LD50 Oral                | LD50 Dermal | LC50 Inhalation              |
|------------------------------------|--------------------------|-------------|------------------------------|
| Manganese(II) sulfate tetrahydrate | 2150 mg/kg (Rat)         |             |                              |
| Manganese(II) sulfate              | LD50 = 782 mg/kg ( Rat ) |             | LC50 > 4.45 mg/L ( Rat ) 4 h |

## (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

# SAFETY DATA SHEET

## Manganese(II) sulfate tetrahydrate

- (c) serious eye damage/irritation; Category 1
- (d) respiratory or skin sensitization;  
Respiratory Based on available data, the classification criteria are not met  
Skin Based on available data, the classification criteria are not met
- (e) germ cell mutagenicity; Based on available data, the classification criteria are not met
- (f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product
- (g) reproductive toxicity; Based on available data, the classification criteria are not met
- (h) STOT-single exposure; Based on available data, the classification criteria are not met
- (i) STOT-repeated exposure; Category 2  
Route of exposure Inhalation  
Target Organs Brain.
- (j) aspiration hazard; Not applicable  
Solid
- Symptoms / effects, both acute and delayed No information available

### SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component                          | Freshwater Fish    | Water Flea        | Freshwater Algae | Microtox |
|------------------------------------|--------------------|-------------------|------------------|----------|
| Manganese(II) sulfate tetrahydrate | 30.6 mg/l LC50 96h | 8.3 mg/l EC50 48h |                  |          |

- Persistence and Degradability**  
Persistence Soluble in water, Persistence is unlikely, based on information available.  
Degradability Not relevant for inorganic substances.  
Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.
- Bioaccumulative Potential** Bioaccumulation is unlikely
- Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils
- Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## SECTION 14. TRANSPORT INFORMATION

**Road and Rail Transport**

**UN-No** UN3077  
**Proper Shipping Name** Environmentally hazardous substances, solid, n.o.s.  
**Technical Shipping Name** Manganese (II) sulfate  
**Hazard Class** 9  
**Packing Group** III

**IMDG/IMO**

**UN-No** UN3077  
**Proper Shipping Name** Environmentally hazardous substances, solid, n.o.s.  
**Technical Shipping Name** Manganese (II) sulfate  
**Hazard Class** 9  
**Packing Group** III

**IATA**

**UN-No** UN3077  
**Proper Shipping Name** Environmentally hazardous substances, solid, n.o.s.  
**Technical Shipping Name** Manganese (II) sulfate  
**Hazard Class** 9  
**Packing Group** III

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                          | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Manganese(II) sulfate tetrahydrate | -                                                   | -                                       | X    | X     | -         | -    | -   | -     | -    |      | X    | -        |
| Manganese(II) sulfate              | -                                                   | -                                       | X    | X     | 232-089-9 | X    | X   | X     | X    | X    | X    | KE-23032 |

**National Regulations**

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 12-Dec-1997  
**Revision Date** 13-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**  
Chemical incident response training.

Legend

|                                                                                                                              |                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>CAS</b> - Chemical Abstracts Service                                                                                      | <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                   |
| <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances | <b>DSL/NDSL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                  |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                                                    | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                       |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                                             | <b>AICS</b> - Australian Inventory of Chemical Substances                                         |
| <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                 |
| <b>WEL</b> - Workplace Exposure Limit                                                                                        | <b>TWA</b> - Time Weighted Average                                                                |
| <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     | <b>IARC</b> - International Agency for Research on Cancer                                         |
| <b>DNEL</b> - Derived No Effect Level                                                                                        | <b>PNEC</b> - Predicted No Effect Concentration                                                   |
| <b>RPE</b> - Respiratory Protective Equipment                                                                                | <b>LD50</b> - Lethal Dose 50%                                                                     |
| <b>LC50</b> - Lethal Concentration 50%                                                                                       | <b>EC50</b> - Effective Concentration 50%                                                         |
| <b>NOEC</b> - No Observed Effect Concentration                                                                               | <b>POW</b> - Partition coefficient Octanol:Water                                                  |
| <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              | <b>vPvB</b> - very Persistent, very Bioaccumulative                                               |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association                         | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code |
| <b>ADR</b> - European Agreement Concerning the International Carriage of Dangerous Goods by Road                             | <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships               |
| <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         | <b>ATE</b> - Acute Toxicity Estimate                                                              |
| <b>BCF</b> - Bioconcentration factor                                                                                         | <b>VOC</b> - (Volatile Organic Compound)                                                          |

**Key literature references and sources for data**  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**